

SHERPA FIRE STOP 2.5

THE INNOVATIVE SOLUTION FOR FIRE PROTECTION

GUARANTEED UP TO 60 MIN. FIRE RESISTANCE!

Wood is becoming increasingly relevant as a building material for use in residential and commercial property as well as in civil engineering. However, as far as fire safety in wooden structures is concerned, there are some obstacles associated with standards and the authorities which still have to be overcome.

SHERPA has developed the world's first standardized connection system, which has undergone 30 and 60 minute fire tests under a static load. SHERPA is therefore a pioneer in the area of fire protection for main and secondary beam connectors.

In particular, SHERPA has investigated the effect of the joint design of a main/secondary beam connection in the case of fire load in a research project implemented in collaboration with Holzforschung Austria (HFA). These tests took place at the Institute for Fire Protection and Safety Research (IBS) in Linz. The goal was to develop cost-effective design variants for a 30 and 60 minute fire load without additional panelling or cladding. The main focus was on the practical viability from an economic perspective.

Dr. Christoph Hackspiel of Holzforschung Austria presented the research results in a paper at the 6th Holz-BauSpezial trade fair in Bad Wörishofen.



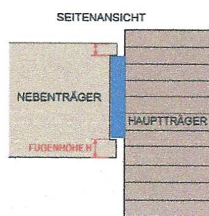
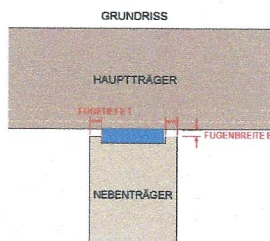
In particular, the fire protection laminate Fire Stop 2.5 from SHERPA proved its mettle in the fire tests. The self-adhesive fire protection laminate which expands in the case of fire is extremely reliable and is easy to process. On the basis of the weather resistance test carried out at the Federal Institution for Materials Testing (BAM) in Berlin, even prolonged installation periods will not present any problems.

The advantages:

- ✓ Highly effective and resistant to ageing
- ✓ Impervious to moisture
- ✓ Free of organic solvents
- ✓ Paintable

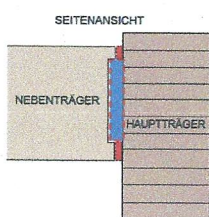
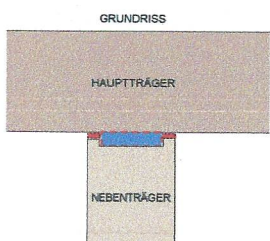
You will find details of the fire test in the conference volume "Brandverhalten der Fuge bei Haupt- und Nebentragwerken" [Fire behaviour of the joint in main and secondary supporting structures]. You can download this from our website www.sherpa-connector.com or order it at office@sherpa-connector.com.





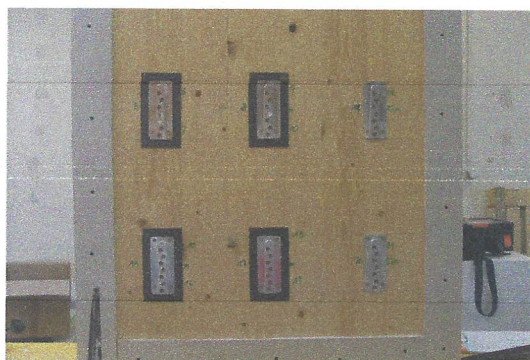
Variante
1

Practical shadow gap up to 5 mm for R30 and R60

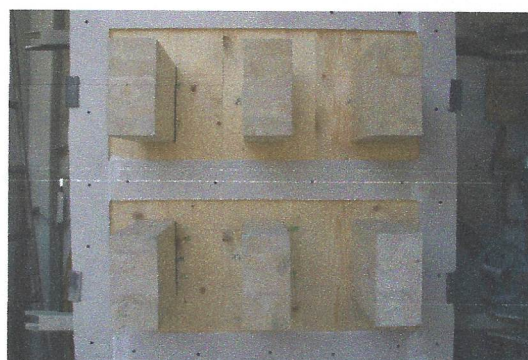


Variante
2

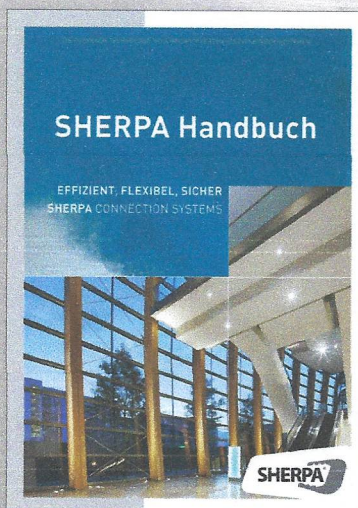
Cost-effective split or flush joint with Fire Stop 2.5 for R30 and R60



Vorbereitung des Prüfkörpers



Prüfkörper eingespannt in den Brandofen



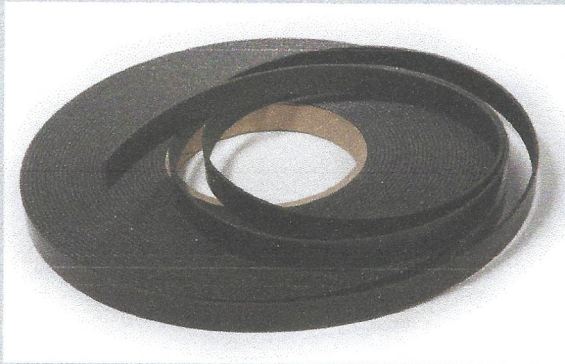
SHERPA manual

The manual offers comprehensive technical data and explanations such as sample calculations and serves as indispensable reference book for processors and decision makers in the planning process of load carrying constructions. The focus in the design of the manual was deliberately put on hands-on topics such as the connections situations of timber to steel or armoured concrete elements.

You can have a look at or download the SHERPA manual from our website www.sherpa-connector.com or you can order a book at office@sherpa-connector.com.



SHERPA Fire Stop 2.5



SHERPA Fire Stop 2.5 is a graphite based intumescent seal with a starting temperature of approx. 150°C (302°F).

- high-impact and resistant to aging
- resistant to humidity influences
- free of organic solvents
- paintable (Preliminary tests absolutely necessary)

Colour	anthracite grey
Texture	highly flexible seal
Spec. weight	1.0 ± 0.2 g/cm ³
Mass per unit area	2.5 mm thickness: 2.5 ± 0.3 kg/m ²
Reaction to fire	B1, difficult to ignite
Expansion temperature	approx. 150°C (305°F)
Expansion ratio*	min. 1 : 7 [30 min/300°C (572°F)]
Expansion pressure*	min. 0.6 N/mm ² [300°C (572°F)]
Thorn flexibility	75 mm [20°C (68°F)]
Humidity resistance	insoluble / hygroscopic (no effect on fire performance)
Substrate	glass fibre web

* Above indicated data of expansion ratio and pressure are taken from the DIBT building material approval. The actual expansion ratio as well as the generated expansion pressure are essentially depending on the fire test of the assembly and also on the joint dimensions of the construction itself.

TEST CERTIFICATE / APPROVAL

- DIBT Z-19.11-80
- UL R 19385

STORAGE REQUIREMENTS

- Store in cool and dry conditions
- Protect from frost and heat
- Storable up to 12 months

Verkaufsdaten	
Bezeichnung	Fire Stop 2,5
Art. Nr.	1000025530
EAN	912004374136
Rollenlänge [m]	25
Breite / Dicke [mm]	20 / 2,5
Stk. / Verpackung	5
lagernd	Ja

SHERPA Fire Stop 2.5 - installation instructions

1

CLEANLINESS

Application should generally be done in separate areas. High levels of dust in the ambient air must be avoided.

2

SURFACES

The bonding surface must be dry and free of dust and greasy residues. Smooth surfaces with high surface energy like wood and metals are unproblematic for bonding. For bonding to surfaces that have been coated or painted prior to the installation, we highly recommend some initial bonding tests.

3

TEMPERATUR

The installation has to be done at temperatures between 15°C (59°F) and 30°C (86°F) for both the ambient temperature and the temperature of the wood. Condensation on the surface of the wood is not permitted. Temperatures in excess of 40°C (104°F) will cause "material shock" and reduce the adhesion. Temperatures between -20°C (-4°F) and +70°C (158°F) are permitted after adhesion/installation

4

MATERIAL CONTROL

Prior to installation, the SHERPA Fire Stop 2.5 has to be checked for defects and damage to insure its proper condition. Rolls that have signs of kinks, detachment of the protective film or any other forms of irregularities should be discarded. The adhesive surface of the SHERPA Fire Stop should not be touched with bare hands.

5

CONTACT PRESSURE

SHERPA Fire Stop only develops its full bonding strength if its entire surface is pressed down "by thumb" (1-2 kg/m²) against the bonding surface. The use of a hard plastic roller eliminates insufficient pressure at certain spots.

6

FULL/FINAL STRENGTH

SHERPA Fire Stop has sufficient initial adhesion to common substrates. The final strength of the adhesive however will only be reached after about 3 days. Under no circumstances may Fire Stop be adhered, then removed and re-adhered!

7

STRESSES AND STRAINS

Unnecessary strains during application such as mechanical shear, tensile stresses as well as deformation of the Fire Stop strip must be avoided..

8

STORAGE

The original packaging protects from light and dust. The storage of SHERPA Fire Stop for up to 12 months is possible at a room temperature of about 23°C (73°F) and a relative humidity of about 55%.

